Now,

$$\begin{aligned} P(k+1) &= 3(k+1)^2 = 3k^2+6k+3 \\ &= P(k)+6k+3 \\ &> (k+1)^2+6k+3 \\ &= k^2+8k+4 \\ &= k^2+4k+4+4k \\ &= (k+2)^2+4k \\ &> (k+2)^2 \text{ since } k>0. \end{aligned}$$

This is the statement $P(k+1)$. The validity of $P(k+1)$ follows from that of $P(k)$. Therefore by the principle of mathematical induction, for all $n \geq 2$, $3n^2 > (n+1)^2$.

Example 4.68 Using the Mathematical induction, show that for any integer

$$n \geq 2, \quad 3^n > n^2$$

Solution:
Let $P(n)$ be the statement that $3^n > n^2$ with $n \geq 2$. Therefore the first stage is $n = 2$. Now, $P(2) = 3^2 = 9$ and $2^2 = 4$. As $9 > 4$, we get $P(2)$ is true
We assume that $P(n)$ is true for $n = k$. That is $P(k) > k^2$. Now,

$$\begin{aligned} P(k+1) &= 3^{k+1} = 3 \times 3^k = 3 \times P(k) \\ &> 3k^2 \\ &> (k+1)^2, \text{ by Example 4.67.} \end{aligned}$$

Hence, for any integer $n \geq 2$, $3^n > n^2$.

Example 4.69 By the principle of mathematical induction, prove that, for $n \in \mathbb{N}$,

$$\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \cdots + \cos(\alpha+(n-1)\beta) = \cos\left(\alpha + \frac{(n-1)\beta}{2}\right) \times \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$$

Solution:
Let $P(n) := \cos(\alpha) + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + ... + \cos(\alpha+(n-1)\beta)$. Then,

$$P(1) = \cos(\alpha) = \frac{\cos(\alpha).\sin(\frac{\beta}{2})}{\sin(\frac{\beta}{2})},$$

which shows $P(1)$ is true. We now assume that $P(n)$ is true for $n = k$. That is,

$$\cos(\alpha) + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \cdots + \cos(\alpha+(k-1)\beta) = \cos\left(\alpha + \frac{(k-1)\beta}{2}\right) \times \frac{\sin\left(\frac{k\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}.$$

We need to prove $P(k+1)$ is true. Now,

$$P(k+1) = \underbrace{\cos(\alpha) + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \cdots + \cos(\alpha+(k-1)\beta)} + \cos(\alpha+k\beta)$$